

LifeTone: Personalized Skin Care Analysis at your Fingertips

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Abstract

In this paper we present LifeTone, a mobile application that aims to offer a luxury good, skincare advice, to a wider audience. Professional skincare consultations can cost hundreds or even thousands of dollars, placing expert advice out of reach for many. Individuals often turn to the internet while seeking an alternative source for advice. There, they find an overwhelming amount of potentially conflicting advice, none of which is tailored to their skin. This process of trial and error not only wastes money on products that might never be used again, but also costs valuable time that could have been spent enjoying healthier skin. LifeTone aims to provide our users with personalized skin care recommendations and instant life-style feedback by analyzing their personal information and facial image. Using those two pools of data, LifeTone recommends a curated list of highly rated products that are appropriate for their skin condition. We present the interface and architecture of LifeTone, and evaluate a context-based and a vectorized model for our product recommender system.

1. Introduction

Routines and rhythms provide a sense of structure and well-being to individuals. Whether it is a post-work gym session, a restorative afternoon nap, or a weekly cleaning ritual, these habitual patterns help individuals maintain a sense of balance and control in their daily lives. Skincare has also become an essential part of self-care, as a broader cultural shift places growing emphasis on personal wellness. One specific instance of this trend is the booming skincare market. Skincare comes in many forms - from makeup to serums and lotions to facemasks - all ending with a shared goal: gaining and maintaining healthy and clear skin.

As more individuals prioritize skincare in their daily routines, the industry continues to expand the overwhelming array of products. Vendors offer various solutions for each problem, with multiple products aimed to treat the same concern, all while promising the perfect remedy. However, the saturation of the market leaves

individuals confused and overwhelmed by the abundance of options available. Furthermore, each individual's skin varies from person to person. A skincare solution that works for one individual may not work for another - even potentially causing harm. Moreover, skincare concerns vary: while one individual struggles with dry or sensitive skin, another could have conditions such as eczema or psoriasis. Although the treatment for these two concerns can be similar, the ability to discern slight differences between the two can help individuals make informed decisions. The overwhelming options lead to decision paralysis, deterring consumers from establishing a skincare routine despite their desire to do so. We tackle this challenge through our AI-powered app, LifeTone.

The primary goal of our work is to help skincare consumers achieve healthy skin by displaying personalized recommendations quickly and pain-free. Our app provides tailored product recommendations given the user's skin type as well as planned out examples of morning and evening routines. By offering a wide array of options to the user with guidance on the best products for their specific needs, we intend to simplify the process of building a skincare routine. LifeTone's goal is to empower users to establish routines that not only nourish their skin but also foster long-term healthy skincare routine and habits.

The rest of the paper is organized as follows: Section 2 reviews related work and market products, Section 3 outlines LifeTone's interface, Section 4 details the recommender system, and Section 5 concludes.

2. Related Work

The integration of machine learning (ML) and artificial intelligence (AI) into healthcare has advanced significantly, particularly in personalized skin care applications. Ethical considerations hold significant weight: Char et al. [1] highlight the risk of algorithms perpetuating human biases, especially racial biases due to underrepresentation in genetic studies. Similarly, Chen et al. [2] emphasize the importance of addressing biases in health records to prevent the marginalization of certain groups. These concerns guide our project's commitment

to ethical AI practices and inclusive data representation.

From a technological perspective, various ML approaches have been explored for healthcare applications. Dambhare and Kumar [3] review algorithms such as decision trees, random forests, support vector machines, and neural networks, illustrating their use in disease prediction and image analysis. Hunter [4] examines the application of augmented reality (AR) in diagnostics to enhance imaging techniques. These methodologies influence our project's use of advanced ML models and the potential integration of AR for improved facial analysis.

In the world of personalized skincare, companies like Atolla and RoC® Skincare have developed AI-powered solutions. Atolla utilizes ML to tailor skincare products to individual needs [5], while RoC® Skincare's "AI Skin Insight™" offers self-diagnostic tools and personalized recommendations [6]. Additionally, Rajegowda et al. [7] present an AI-assisted skincare recommendation system to reduce the need for expert consultation. Our work diverges from these efforts by incorporating personalized lifestyle advice and providing product recommendations. We contribute a novel approach to AI-powered skincare solutions by evaluating two distinct ML models that enhance recommendation accuracy and scalability.

3. System Design

The LifeTone mobile application was designed with a modular and scalable architecture, ensuring efficiency, maintainability, and seamless user interaction. Users are greeted with a minimalistic and intuitive interface to enhance approachability and accessibility with a visually appealing aesthetic. The system is structured into two primary components: the frontend—responsible for user interface and interaction, and the backend—handles data processing, authentication, and API integrations.

3.1 Frontend & UI/UX Design

The frontend of LifeTone is built using React Native, enabling cross-platform compatibility for both iOS and Android. Development was enhanced with Expo Go for rapid prototyping and testing. User authentication is secured via Supabase Auths. Axios handles HTTP communication with the backend. It also integrates the Face++ API for real-time image capture and analysis, and delivers AI-driven skincare recommendations based on data processed by Google Gemini AI.

The UI/UX design is drawn on Figma due to its diverse toolsets and collaboration options. Color schemes and typography are carefully chosen to reflect the leading nature of our work: student-driven, new ideas, and an ambitious goal—simplifying entry into skincare for the general public while shedding light on underrepresented skin types. All recommendations, such as lifestyle tips,

ingredients, and product recommendations, are tailored towards each user using current skin conditions, skin allergies, and lifestyle. The onboarding sequence introduces LifeTone's features and benefits through a welcome screen. Users complete the sign-up/login process via manual entry. An initial profile setup collects key skincare information to tailor recommendations.

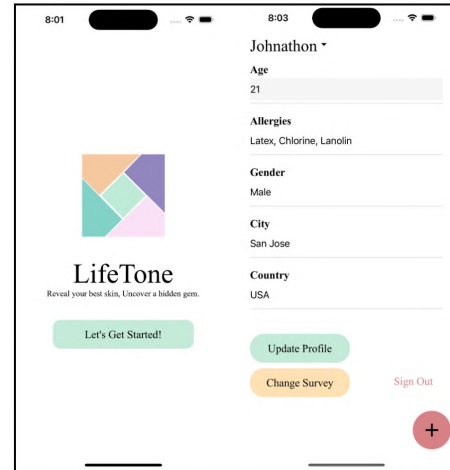


Figure 1. Welcome and Profile Screens

Lifetone's application provides an image upload interface for AI-driven skin analysis. It utilizes real-time image validation with Face Analyze API from Face++. A secure image validation system prevents improper or low-quality submissions. The application delivers a personalized skincare recommendations display, utilizing a card-based UI for intuitive browsing.

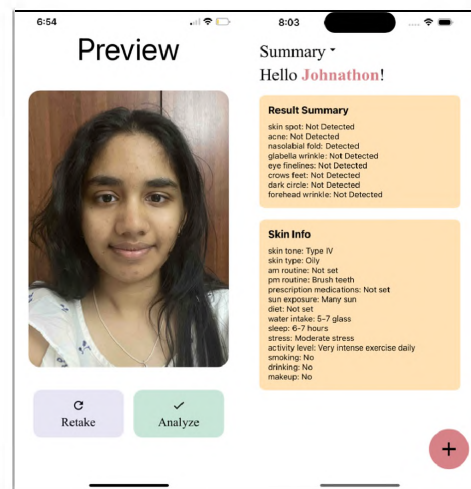


Figure 2. Image Upload and Skin Analysis Home Screens

Users can interact with suggested skincare routines, product recommendations, and educational resources through swipe gestures. Integration with GeminiAI enables the utilization of skincare data to map out lifestyle tips and other individual ingredients to look out for, and the recommender system suggests skincare

products on a separate page. Overall, the structured design promotes ease of use, AI-powered customization, and a user-friendly experience rooted in inclusivity and innovation.

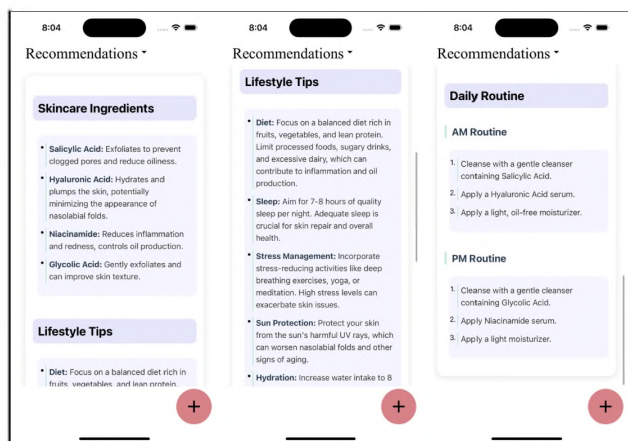


Figure 3. Skincare Recommendations Screen

3.2. Backend

The LifeTone mobile application's backend is built on a multi-tiered architecture using Node.js with Express.js for efficient, non-blocking request handling and routing. Data storage is managed via Supabase with PostgreSQL, using the ORM tool, Prisma, for streamlined relational interactions. Secure user authentication is reinforced through OAuth2. A dedicated Python module developed with FastAPI, Scikit-learn, and Pandas delivers AI-driven skincare recommendations. The service is deployed via Uvicorn to improve concurrency.

Ngrok facilitates secure remote debugging during development. Image validation middleware is integrated with OpenCV and TensorFlow.js to ensure high-quality image input. System performance and user engagement are continuously monitored with Google Analytics, contributing to the app's scalability with proactive system tuning and enhancements. Backend and frontend development is coordinated via GitHub, with model prototyping and data exploration using Google Collab. Collectively, these components offer a secure, efficient, and scalable backend infrastructure for LifeTone's innovative skincare recommendation service.

4. Recommender System

Two models were evaluated to determine the optimal approach for recommending cosmetics: a context-based model (Model 1) and a vectorized model (Model 2). The evaluation is based on key dimensions such as ingredient matching, filtering logic, similarity measurement, computational complexity, scalability, and several quantitative metrics: accuracy, precision, recall, and runtime performance.

The dataset used for both of the models was compiled from two publicly available Kaggle sources on skincare products.^{1 2}

4.1. Filtering Logic and Similarity Measurement

Our dataset consists of 1,472 cosmetic product records, each with detailed attributes including Label, Brand, Name, Price, Rank, Ingredients, and skin type indicators for Combination, Dry, Normal, Oily, and Sensitive. Both models begin by filtering products based on user-defined criteria such as skin type, product label, brand, and price range. Based on simple attribute comparisons, this initial filtering stage reduces the candidate set and improves overall efficiency. However, the two methods diverge in how they rank the filtered candidates:

Model 1 applies a rule-based matching approach, where the binary or weighted vectors are compared using basic similarity measures. A binary vector is a sequence where each position represents the presence, denoted as 1, or absence, denoted as 0, of a specific ingredient. In the weighted vector, certain key ingredients receive higher numerical values. To measure for similarity, Model 1 uses Jaccard similarity, which is defined as the ratio of shared ingredients to the total number of unique ingredients in two products. This method infers similarity from the presence or absence of keywords and does not leverage deeper numerical similarities, unless manually weighted.

Model 2 refines the candidate list by computing cosine similarity between Term Frequency-Inverse Document Frequency (TF-IDF) vectors. Cosine similarity measures the angular distance between two vectors in the high-dimensional space, providing a quantitative basis for ranking recommendations. This metric is particularly effective for text data and is well-suited to capturing subtle ingredient relationships in a data-driven manner.

While both models employ a form of ingredient vectorization, Model 1 relies on manually assigned ingredient importance through weighted binary vectors, whereas Model 2 generates statistical weights automatically via TF-IDF.

4.2. Context-Based Model (Model 1)

The context-based model relies on a traditional one-hot encoding technique to capture the presence or absence of ingredients within cosmetic products. In this approach, each product is represented by a binary vector indicating whether specific ingredients appear in its formulation.

Moreover, this model incorporates a layer of domain-specific knowledge through contextual weighting rules. Certain deemed critical ingredients, such as

¹ www.kaggle.com/eward96/skincare-products-and-their-ingredients

² www.kaggle.com/datasets/kingabzpro/cosmetics-datasets

titanium dioxide, retinol, or ascorbic acid, are assigned higher weights to emphasize their importance in skincare applications. The model then uses these binary or weighted representations in a similarity computation framework. However, this reliance on simple string matching leads to a relatively coarse and less nuanced set of recommendations. Hence, Model 1 is straightforward and computationally efficient, requiring approximately 0.051 seconds per recommendation query. It achieves lower performance levels in terms of accuracy (approximately 75%), precision (approximately 70%), and recall (approximately 65%). Furthermore, the manual filtering and binary comparison approach may not scale well when handling large volumes of complex data. Incorporating fuzzy matching or probabilistic weighting would improve the recognition of synonyms and nuances.

4.3. Vectorized Model (Model 2)

The vectorized model employs advanced natural language processing techniques to generate numerical feature representations of product ingredients. This approach utilizes TF-IDF vectorization to convert ingredient data into dense, informative feature vectors. The model then applies cosine similarity as a metric to quantitatively evaluate the similarity between the feature vectors of different products. This process captures the relative importance of various ingredients and provides a more granular assessment of product similarity. Although this method is more computationally intensive, averaging around 0.128 seconds per query, it offers significantly improved performance metrics. The model demonstrates an accuracy of approximately 89%, a precision of 85%, and a recall of 82%. These improvements are attributed to the TF-IDF methodology's ability to discern subtle differences in ingredient profiles and properly weigh the influence of key ingredients. This allows the model to recognize, for instance, that 'ascorbic acid' and 'Vitamin C' are semantically equivalent even when expressed differently. Moreover, the vectorized model would scale effectively with larger datasets, thanks to efficient matrix operations inherent in the TF-IDF computation. In addition, exploring advanced embedding methods such as BERT or Word2Vec could further enhance semantic understanding and capture context-dependent nuances.

4.4. Computational Complexity and Scalability

The context-based model is computationally efficient due to its reliance on simple string matching and binary arithmetic, resulting in fast runtimes (approximately 0.051 seconds per query). This translates to a time complexity of $O(n)$, where n is the number of ingredients or product features being matched. However, this simplicity comes at the cost of reducing the fidelity of

ingredient matching. In contrast, the vectorized model incurs higher computational costs, averaging around 0.128 seconds per query. TF-IDF vectorization and cosine similarity computations are more demanding with a time complexity of $O(n \log n)$. However, modern hardware and optimized linear algebra libraries mitigate the costs. Thus, the trade-off is acceptable, given the improvement in recommendation quality.

Regarding scalability, Model 1's manual filtering and one-hot encoding approach can become a bottleneck when processing large datasets with space complexity of $O(n)$ due to dense binary matrix representations. Binary matrices often grow in size and sparsity, thus impairing performance. On the other hand, Model 2, with its reliance on sparse matrix representations and efficient TF-IDF operations, is better suited for scaling up to larger product catalogs with space complexity of $O(n \log n)$ in typical implementations. This scalability is critical for real-world applications where the volume and variety of cosmetic products are continually expanding.

4.5. Quantitative Performance Metrics

Table 1. Quantitative Performance – Model 1: Context-Based vs Model 2: Vectorized.

	Model 1	Model 2
Accuracy	75%	89%
Precision	70%	85%
Recall	65%	82%
Runtime	0.051 seconds	0.128 seconds

For model evaluation, 10% of the data was reserved as a test set. Accuracy, precision, and recall are measured by comparing the recommended cosmetic products against a curated set of relevant items for each query, where accuracy is defined as the proportion of correct recommendations (i.e., both true positives and true negatives) over the total number of items, precision represents the fraction of recommended items that are truly relevant (true positives divided by the sum of true positives and false positives), and recall quantifies the proportion of all relevant items that were successfully recommended (true positives divided by the sum of true positives and false negatives). For each query, the recommendations produced by both models are compared against a curated list of relevant cosmetic products. This curated set serves as the "ground truth" for judging which recommendations are correct.

Referring to Table 1, our context-based model, using frequency and priority-based weighting, had a lower overall accuracy, precision, and recall than our vectorized model, which employed TF-IDF vectorization and cosine similarity. In return, however, given the context and small dataset, it achieved higher speeds. These metrics indicate that while the context-based model is faster, its simplistic

nature leads to poorer overall performance. In contrast, despite a modest increase in computational overhead, the vectorized model provides significantly enhanced recommendations with better accuracy, precision, and recall.

4.6. Model Analysis Conclusion

Consider a user with dry skin seeking naturally formulated cosmetic products. The context-based filtering model (Model 1) identifies such products by scanning for explicit keywords (i.e., “dry”, and “natural”) in product descriptions. While this approach captures items that overtly mention these terms, it can overlook products that use alternative expressions (e.g., “botanical extracts”) to describe natural ingredients. In contrast, the vectorized model (Model 2) transforms texts into TF-IDF feature vectors and employs cosine similarity to capture direct and indirect references to desired attributes.

Based on our findings, the vectorized model is adopted as the primary recommendation engine because it captures complex ingredient relationships and produces higher relevant results. Moreover, similar innovations have been observed in Subhan et al. (2025), where a Neural Network Collaborative Filtering (NNCF) method in e-commerce overcomes the cold start problem, achieving a recall of 0.74 and outperforming several traditional approaches. [8]

Nevertheless, we propose a hybrid strategy for further optimization. This approach uses the context-based filter as an initial, fast screening tool that narrows the candidate product list, after which the vectorized model refines the shortlist with an in-depth, nuanced analysis. This strategy could reduce the overall computational overhead and maintain the high recommendation quality observed with the vectorized model.

5. Conclusions

The LifeTone mobile application successfully delivers personalized skincare recommendations through the integration of AI and machine learning. Two models were evaluated: a context-based filtering model and a vectorized model. While the context-based model offered faster runtimes, it was less accurate than the vectorized model, achieving 75% accuracy compared to the 89% accuracy of the TF-IDF approach, albeit with a longer runtime (Table 1). These results highlight the vectorized model’s ability to capture nuanced ingredient relationships and deliver more relevant recommendations.

The application’s user-friendly design, developed with React Native and integrated with tools like Face++ API and GeminiAI, ensures a seamless and engaging user experience. By supporting personalized skincare routines and lifestyle tips, LifeTone achieves its goal of making skincare accessible to a broader audience.

For future work, implementing a hybrid approach, combining the fast filtering of the context-based model with the nuanced analysis of the vectorized model, could optimize both accuracy and efficiency. Additionally, expanding the dataset to include a wider variety of skincare products, integrating multilingual support, and enhancing real-time skin condition monitoring are potential areas for further improvement.

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